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ORIGINAL ARTICLE

MLDC: multi-lung disease classification using quantum classifier and artificial neural networks

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Abstract
Lung diseases are one of the most common diseases around the world. The risk of these diseases are more in under-developed and developing countries, where millions of people are battling with poverty and living in polluted air. Chest X-Ray images are helpful screening tool for lung disease detection. However, disease diagnosis requires expert medical professionals. Furthermore, in developing and under-developed nations, the doctor-to-patient ratio is comparatively poor. Deep learning algorithms have recently demonstrated promise in the analysis of medical images and the discovery of patterns. In this current work, we have proposed a model MLDC (Multi-Lung Disease Classification) to detect common lung diseases. It introduces a MLDC feature extraction model with two different new classifiers, considering ANN (an artificial neural network) and QC (a quantum classifier). In this proposed model, tests are performed on the LDD (Lung Disease Dataset), which includes COVID-19, pneumonia, tuberculosis, and a healthy person's lung from chest X-ray images. Our proposed model achieves an accuracy of 95.6% for MLDC-ANN and 97.5% for MLDC-QC at a lower computational cost.

Keywords COVID-19 · Pneumonia · Tuberculosis · Image processing · Deep learning · Machine learning · Quantum classifier

1 Introduction

Millions of individuals worldwide suffer from lung diseases such as asthma, lung cancer, pulmonary fibrosis, and pulmonary disease, making lung diseases a serious public health concern. The severity of these illnesses can vary greatly, with some people only showing minor symptoms that barely affect their daily activities while others may experience severe symptoms that necessitate intensive medical care and support. The primary reasons for the disease are, the person's general health and age,environmental factors like exposure to pollutants or allergies, and smoking. COVID-19 pandemic exaggerated the existing problems and added new challenges. Chest X-Rays, CT Scans and MRIs are common imaging techniques used for lung disease diagnosis. Out of the three, Chest X-Rays are mostly used because they are comparatively less expensive and amount of radiation done to get chest images is low.

Deep learning or machine learning models in addition to it, has made it easier to detect the early stage features from the X-Ray images. For the applications involving classification of medical images, deep learning has been a promising and effective approach. It has encouraged the creation of intelligent systems that can help medical professionals to make better decisions regarding patient's health. According to a US study on the use of Artificial Intelligence (AI) in medical image analysis, a deep learning algorithm can spot anomalies on Chest X-Rays images equivalent to that of a professional radiologists [1]. Thus, it can help medical professionals in the easy diagnosis of diseases.

Our research contributes to build a model to detect various lung diseases using deep learning techniques. Deep learning is pivotal in X-ray image classification due to its

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unparalleled ability to automatically discern complex patterns. Specifically, Convolutional Neural Networks (CNNs) excel at this task by hierarchically extracting features from images, crucial in detecting anomalies of varying scales. Unlike traditional methods, deep learning models learn directly from raw data, eliminating the need for manual feature extraction. Moreover, they can adapt to new data and benefit from transfer learning, even with limited datasets. Additionally, deep learning enables ensemble approaches for enhanced performance and offers visualization techniques to provide insights to healthcare professionals. Ultimately, it accelerates the diagnostic process, potentially reducing human error and enhancing the efficiency of healthcare delivery.

1.1 Motivation

Motivation for developing the proposed model are listed as follows:

- Improve the Early Detection and Diagnosis of respiratory diseases.
- Reduce the burden on the medical professionals by providing a tool that can help in quick diagnosis of patients with lung diseases.
- Maximize the utility of Chest X-rays, ensuring high accuracy even in resource-constrained settings.
- Automate the process of feature extraction and analysis to reduce the human error, thereby increasing the overall efficiency.

1.2 Contributions

We have proposed one feature extraction module and two classification modules. Feature Extraction is done using Convolutional Neural Network(CNN) which consists of different convolution layers and max-pooling layers. We have proposed two classifiers in our classification module, the first classifier consists of a fully connected neural network and the second one is a quantum classifier.

- The model focuses on the improvement of unprocessed X-Ray images collected from various datasets.
- In feature extraction module, Multi-Lung Disease Classification-Feature extractor(MLDC-F) we have used CNN for the extraction of infected areas in X-Ray images.
- It further classifies the X-Ray images with the help of two different classifiers proposed in our classification module.
- The first classifier,Multi-Lung Disease Classification-Artificial Neural Network(MLDC-ANN) consist of different fully connected layers.
- The second Classifier, Multi-Lung Disease Classification-Quantum Classifier(MLDC-QC) is MMS(Multi Multi Single) quantum classifier.

2 Related work

The literature has many studies on the identification of lung diseases using deep learning. Most of the existing work focus on binary classification of lung disease and majority of the studies have used chest X-Rays as part of their research [2–6] and [7]. A deep learning-based TB detection technique was proposed by Liu et al. [2]. The scientists proposed a transfer learning-based CNN model that has an accuracy of 85.68%. Dokur et al. [3] developed a classification model based on CNN that classifies X-Ray images of chest with an accuracy of 86% which has been obtained by utilizing 12- class chest X-Ray images dataset in testing. Chouhan et al. [4] conducted a research that classified between pneumonia and normal X-ray pictures using a group of five transfer learning-based models. The model obtained an accuracy of 96.4% after training there hybrid model on ImageNet Dataset. Deep learning, a subset of machine learning, has shown remarkable prowess in handling complex, high-dimensional data. It can be applied effectively in both single and multi-objective optimization tasks [8–10].

A deep learning-based network COVID-Net has been proposed by Wong and Wang [5] to distinguish among Covid-19 and healthy X-Ray images of chest. There are 11 convolution layers in the COVID-Net model. The authors employed a dataset that was a hybrid of Dr. Joseph Cohen's COVID chest X-Ray dataset [11] and the Kaggle dataset [12] of chest X-Rays, which includes multiple classes such as Pneumonia (normal, viral infection, bacterial infection) and COVID-19. The authors received an accuracy score of 83.5%. Maghdid et al. [6] proposed a study in which they used a deep learning-based model and transfer learning approaches to automate the identification of the COVID-19 disease in chest X-Ray image. The authors utilized a simple CNN design in which one convolutional layer with 16 filters was used as an activation function, followed by batch normalization. The AlexNet model was also adjusted for transfer learning. The proposed model was found to be 95.00% accurate.

Tucker and Ghoshal [7] looked at the diagnostic unpredictability and interpretability of deep learning-based algorithms for COVID-19 identification in X-Ray images. To assess uncertainty, they have utilized Bayesian Convolutional Neural Networks (BCNN) to deep learning models. To distinguish between accuracy and uncertainty, they used 70 COVID-19 positive chest X-Ray images from

a dataset made public by Cohen [11] as well as Kaggle's dataset [12]. For preprocessing, all images were reduced to 512×512 pixels. A few research efforts have used Data augmentation for transfer learning. Using the VGG16 deep learning model, the suggested Bayesian inference technique has an accuracy of 92.86% for detection.

Quantum-based machine learning is a study that develops and use quantum software to enable such a machine learning that is faster than classical computers [13]. Nivelkar et al., [14] proposed a model for supervised machine learning using quantum computing. The ever-growing volume of data, current machine learning systems are increasing beyond the limits of conventional computational models. In order to increase the discriminability of the conventional CNN model, Tong et al. [15] investigated the potential of parameterized quantum circuits and proposed a quantum-classical deep network architecture. Chen et al. [16] for binary image classification also proposed two scale-inspired local feature extraction algorithms in Tensorflow quantum framework based on Quantum convolution neural network(QCNN). Quantum algorithms are constructive for enhancing the efficiency of standard algorithms by using the properties of quantum bits. Choudhuri et al. [17] proposed quantum classical convolutional neural network architecture for brain MRI tumor classification that supports the capability of the network to identify and categorize brain tumors with an accuracy rate of 97–98% across many datasets.

The existing research works have focused on binary classification. In the current work we have proposed multi-class classification of lung diseases.

3 Proposed model: MLDC

The proposed model MLDC (Multi-Lung Disease Classification) consists of a feature extraction module and two classification modules. The feature extractor module preprocesses the training data and uses MLDC-F (Multi-Lung Disease Classification-Feature Extractor) to extract features and the second module is for classification.We have proposed two classifiers, MLDC-ANN (Multi-Lung Disease Classification-Artificial Neural Network) a fully connected neural network and MLDC-QC (Multi-Lung Disease Classification-Quantum Classifier) which is a MMS (Multi Multi Single) Quantum Classifier. The overview of MLDC is presented in Fig. 1.

3.1 Proposed feature extraction module

The proposed Feature Extraction Module consists of two major components LDD(Lung Disease Dataset) and MLDC-F.

3.1.1 LDD preprocessing

The existing research work focus on binary classification such as Covid and Non-Covid detection, pneumonia and healthy Chest X-Rays. This necessitates development of a model that can classify among different lung diseases. In order to develop such a model, we need training dataset that consists of chest X-Ray images of different lung diseases. To the best of our knowledge, there is no such multiclass chest X-Ray images for lung diseases. This motivated us to construct a common dataset that can be used for detection of multiple lung diseases. Hence, we have constructed a multiclass lung disease dataset(LDD).

- **Dataset preparation:** Cohen et al. [18] has created a binary class dataset of X-Ray images of chest for Covid-19 and non-Covid-19 Classification. Researchers from all over the world constantly contribute images with this database, which is updated from time to time. For normal and pneumonia X-Ray images dataset we have used another binary class Chest X-ray images(Pneumonia) Dataset [19]. [20] have constructed a binary dataset of X-Ray images of chest for Tuberculosis (TB) positive cases along with Normal images.

  We have combined all these three datasets together and have made four classes out of it namely Covid-19, Tuberculosis, Pneumonia and Normal. We used 523 covid-19 positive X-ray images,1,338 pneumonia positive X-Ray images and 300 tuberculosis positive chest x-ray images. A total of 2,161 X-ray images of a healthy person are taken from the above-mentioned datasets [18–20]. The number of Positive X-ray images and that of a healthy person's X-Ray images are taken in same numbers in order to maintain the uniformity in the data.
- **Dataset preprocessing:** The X-Ray images considered for this experiment were collected from different imaging centres and had been taken through different machines. Because of these differences we have to preprocess the dataset. We have resized the dataset collected from various sources into a common dimension of $224 \times 224 \times 3$. We have also normalized the dataset. Figure 2 shows some examples of X-Ray images from LDD.

3.1.2 MLDC-F

MLDC-F consists of six different blocks. Figure 3 presents the feature extraction part of the proposed model MLDC.

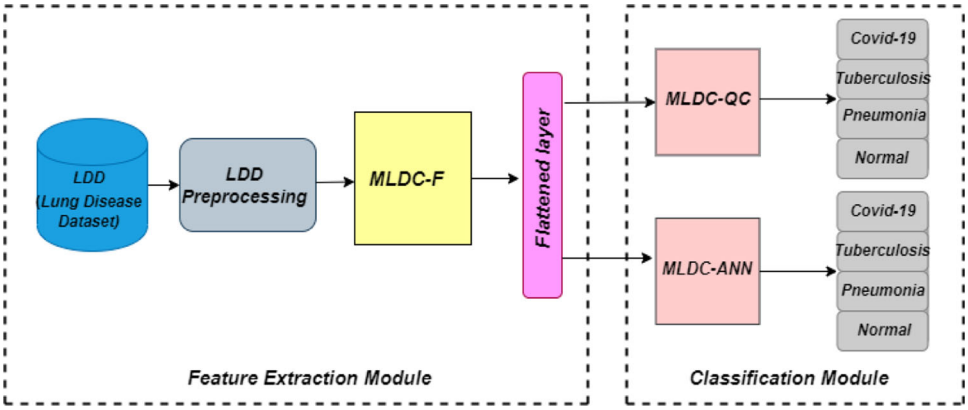

Fig. 1 Proposed MLDC model

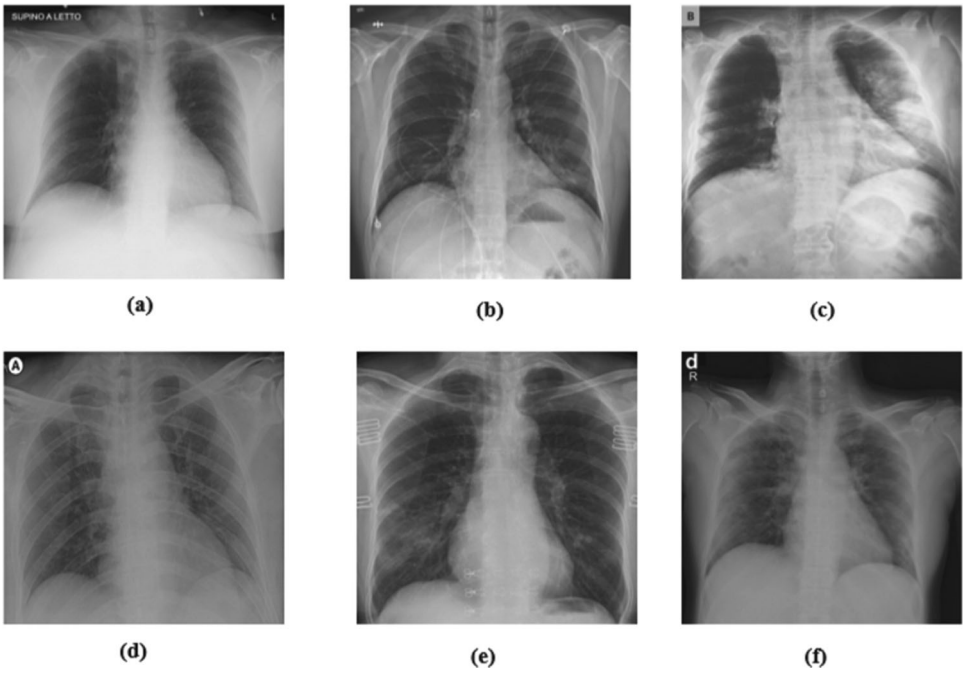

Fig. 2 Sample images from LDD

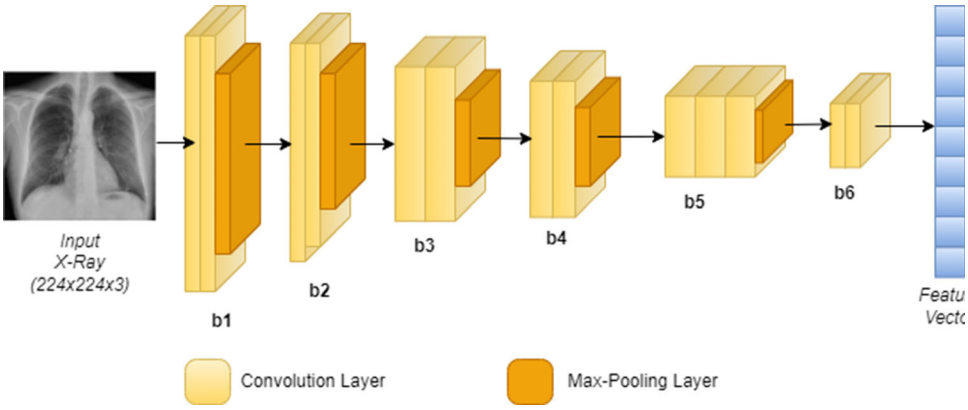

Fig. 3 Proposed feature extractor MLDC-F

Algorithm 1 MLDC-F

Input: LDD - $\{L^1, L^2 ... L^K\}$ ▷ K is number of classes in LDD
Output: F- $\{F^1, F^2 ... F^K\}$ ▷ F is the feature vector for each class K

Initialize:
$N \leftarrow$ Number of Images in each class
$E \leftarrow$ Number of Epochs
for $x \leftarrow 1, K$ **do**
  $N \leftarrow Count(L^K)$
  for $i \leftarrow 1, N$ **do**
    $I \leftarrow L_i^K$
    Resize image I to 224x224x3
    Normalize the image I
    for $e \leftarrow 1, E$ **do**
      $I1 \leftarrow Conv1$(I, 3x3, 8)
      $I2 \leftarrow Conv2$(I1, 3x3, 8)
      $I3 \leftarrow MaxPool1$(I2, 2x2)
      $I4 \leftarrow Conv3$(I3, 3x3, 16)
      $I5 \leftarrow Conv4$(I4, 3x3, 16)
      $I6 \leftarrow MaxPool2$(I5, 2x2)
      $I7 \leftarrow Conv5$(I6, 3x3, 32)
      $I8 \leftarrow Conv6$(I7, 3x3, 32)
      $I9 \leftarrow MaxPool3$(I8,2x2)
      $I10 \leftarrow Conv7$(I9, 3x3, 64)
      $I11 \leftarrow Conv8$(I10, 3x3, 64)
      $I12 \leftarrow MaxPool4$(I11,2x2)
      $I13 \leftarrow Conv9$(I12, 3x3, 128)
      $I14 \leftarrow Conv10$(I13, 3x3, 128)
      $I15 \leftarrow Conv11$(I14, 3x3, 128)
      $I16 \leftarrow MaxPool5$(I15,2x2)
      $I17 \leftarrow Conv9$(I16, 3x3, 256)
      $I18 \leftarrow Conv10$(I17, 3x3, 256)
    end for
  end for
end for
$F^K \leftarrow I18$
return

- **b1**: Block b1 consists of two convolution layers, each with eight filters of size 3×3. We have applied 2×2 max-pooling and stride of 1 after both convolution layers. Block b1 will accept an image of size $224 \times 224 \times 3$. The resultant feature map of the first block b1 will be of size 112×112 after applying both convolution and max-pooling. Block b2 will receive this feature map as an input.
- **b2**: Block b2 consists of two convolution layers, each with sixteen filters of size 3×3. We have applied 2×2 max-pooling and stride of 1 after both convolution layers. The resultant feature map of block b2 will be of size 56×56 after max-pooling. Block b3 will receive this feature map as input.
- **b3**:Block b3 consists of two convolution layers,each with 32 filters of size 3×3. We have used 2×2 max-pooling and stride of 1 after both convolution layers.The resultant feature map of block b3 will be of size 28×28 after max-pooling. Block b4 will receive this feature map as input.
- **b4**:Block b4 consists of two convolution layers,each with 64 filters of size 3×3. We have used 2×2 max-pooling and stride of 1 after both convolution layers. The resultant feature map of block b4 will be of size 14×14 after max-pooling. Block b5 will receive this feature map as input.
- **b5**:Block b5 consists of three convolution layers, each with 128 filters of size 3×3. We have applied 2×2

max-pooling after all three convolution layers. The resultant feature map of block b5 will be of size 7×7 after max-pooling. Block b6 will receive this feature map as an input.

- **b6**:Block b6 consists of two convolution layers, each with 256 filters of size 3×3. We haven't applied max-pooling in this block. Block b6 will accept a feature map from b5.

All the above boxes uses 'ReLu' as their Activation Function. The resultant feature map from block b6 is then flattened to a 1-D Array feature vector so as to apply different classification techniques.The detailed architecture of MLDC-F is shown in Table 1. The detailed explanation of MLDC-F is presented in Algorithm 1.

3.2 Proposed classification module

3.2.1 MLDC-ANN

MLDC-ANN consists of 4 fully connected neural network. Figure 4 represents the classification part of the proposed model called MLDC-ANN. As illustrated it contains 512 neurons in the first fully connected layer (fc1). The fc1's output will be fed into fc2 as an input. fc2 has 512 neurons as well. Similarly, fc2's output will be fed to fc3 and further to fc4, both of which have 128 neurons. The output from fc4 will be fed to the last output layer, and the activation function sigmoid will be used at the output layer to evaluate the four classes of the image and the learning rate

Algorithm 2 MLDC-ANN

Input: F-{F^1,F^2...F^k} ▷ F is the feature vector for each class k
Output: Labelled Class for the corresponding feature vector for the dataset.

$MLDC_ANN \leftarrow F^K$
for $x \leftarrow 1, K$ **do**
  $D \leftarrow Convert_1D_array(F^K)$
  $D1 \leftarrow Dense1(D, 512, ReLu)$
  $D2 \leftarrow Dense2(D1, 512, ReLu)$
  $D3 \leftarrow Dense3(D2, 128, ReLu)$
  $D4 \leftarrow Dense4(D3, 128, ReLu)$
  $Labelled_Class \leftarrow (D4, Softmax)$
end for
return

Table 1 Detailed architecture of MLDC-F

Block	Layer name	Parameters		
Input	Input	Image size: 224x224x3		
b1	Convolution	3×3 Filter size	8 Filters	Relu activation function
	Convolution	3×3 Filter size	8 Filters	Relu activation function
	Pooling	2×2 Pooling size	Stride 1	Max-pooling method
b2	Convolution	3×3 Filter size	16 Filters	Relu activation function
	Convolution	3×3 Filter size	16 Filters	Relu activation function
	Pooling	2×2 Pooling Size	Stride 1	Max-pooling method
b3	Convolution	3×3 Filter size	32 Filters	Relu activation function
	Convolution	3×3 Filter Size	32 Filters	Relu activation function
	Pooling	2×2 Pooling size	Stride 1	Max-pooling method
b4	Convolution	3×3 Filter size	64 Filters	Relu activation function
	Convolution	3×3 Filter size	64 Filters	Relu activation function
	Pooling	2×2 Pooling Size	Stride 1	Max-pooling method
b5	Convolution	3×3 Filter size	128 Filters	Relu activation function
	Convolution	3×3 Filter Size	128 Filters	Relu activation function
	Convolution	3×3 Filter size	128 Filters	Relu activation function
	Pooling	2×2 Pooling size	Stride 1	Max-pooling method
b6	Convolution	3×3 Filter size	256 Filters	Relu activation function
	Convolution	3×3 Filter Size	256 Filters	Relu activation function

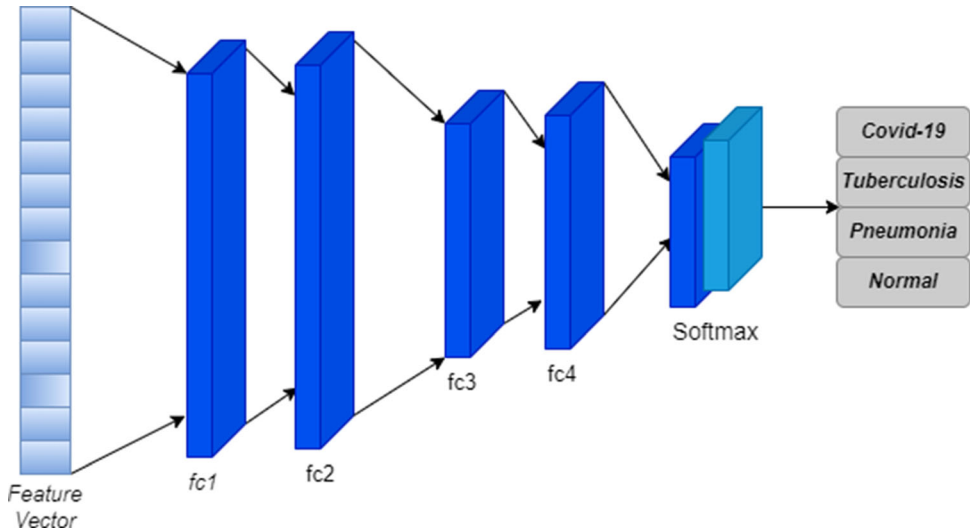

Fig. 4 Proposed classifier MLDC-ANN

is 0.0001. The detailed explanation of MLDC-ANN is presented in Algorithm 2.

3.2.2 MLDC-QC

Quantum classifiers attempt to classify data into meaningful classifications like its classical counterparts. Entanglement, qubit encoding, and a measurement process of the range of possible outcomes are all part of the process. As classifiers, the suggested quantum circuits conduct hidden unitary operations on encoded quantum states, providing the corresponding transformation values of entangled states. In order to train the circuit, a measuring operation must be carried out. These observed probabilities are used as a training dataset, and then tuned to achieve the desired outcome. The proposed classifier MLDC-QC has been presented in Fig. 5.

- **MMS (Multi multi single):** From MMS, we pass the input features to the proposed processing unit (MMS). This classifier has three distinct levels. There are three distinct layers present here: the encoded layer (responsible for encoding classical features into quantum bits or states), the variational circuit layer (which determines the connections between the qubits), and the measurement layer (Converts the qubits back to classical).
- **Encoded layer:** In the Encoded Layer, classical values are converted to quantum states using Hadmard gates. The four blocks of features are encoded into qubits using a series of single rotational gates: RY, RX, texRZ and RY. Each qubit state that has an angle about the X, Y, or Z axes undergoes one of these operations.
- **Circuit layer:** After the state has been encoded successfully, it can be decoded using either a single CNOT gate or a multi-entangled CNOT gate.

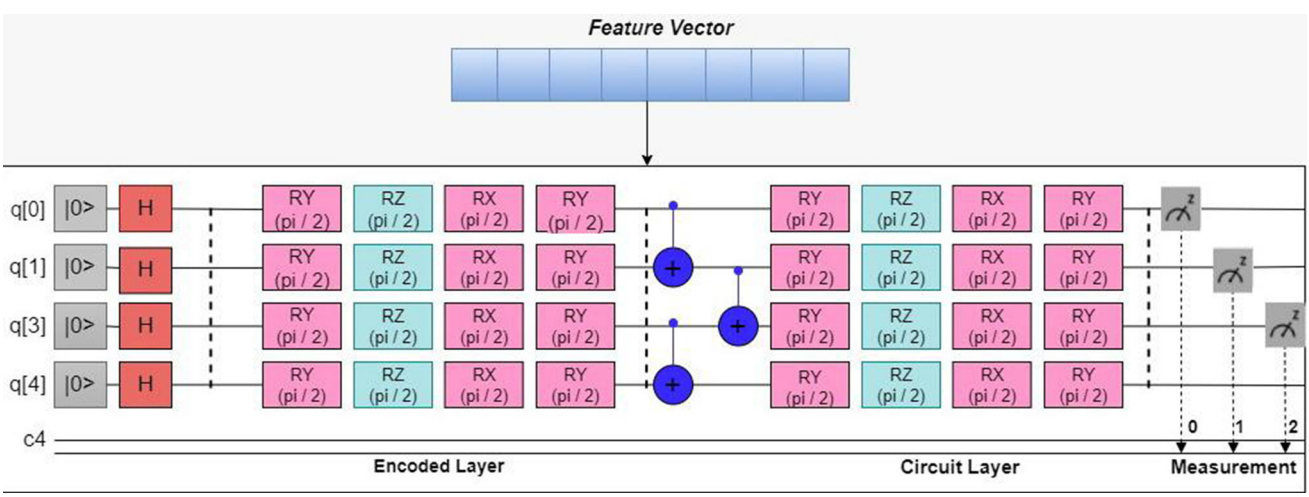

Fig. 5 Proposed classifier MLDC-QC

Entanglement between states in the even and odd quantum numbers modifies the produced Qubits (Encoded Layer Output) from one state to another. In this case, entanglement is achieved through the use of three CNOT gates: two multi-CNOT gates and one single-CNOT gate. The last step is to implement and demonstrate the mapping between entangled quantum states using four rotating gates (RY, RX, RZ, and RY). The layer of variational circuitry is repeatedly trained over and over again throughout the training phase.

- **Measurement layer:** The measurement layer may also be referred to as the decoding phase. The Pauli-Z gate was responsible for converting qubits into the desired output, which consisted of classical values. The detailed explanation of MLDC-QC is presented in Algorithm 3.

Algorithm 3 MLDC-QC

Input: F is re-scaled to element-wise transformation between $[0, 1]$ (denoted as $\mathrm{M}(m^l \in \mathbb{R}^M)$).
Output: Labelled Class for the corresponding feature vector n^l for the dataset.

$\mathcal{L} = (m^l, n^l)_{l=1}^{L}$
Initialize:
$Data_Quantumbits \leftarrow 0\rangle$
$Q \leftarrow Qdepth$
$\delta \leftarrow$ Quantum parameters optimized using Classical computer
$Data_Quantumbits \leftarrow State_Configuration$
$\mathrm{RY}(\mathrm{m}_i^l)\mathrm{RZ}(\mathrm{m}_i^l)\mathrm{RX}(\mathrm{m}_i^l)\mathrm{RY}(\mathrm{m}_i^l))\ \forall i \in\ 1, 2, \ldots, M$
for $i \leftarrow 1, Q$ **do**
    $entanglement$: **CNOT**[0,Data_Quantumbits, 2]followed by;
    $another_shift$: **CNOT**[1,Data_qubits,2];
    $\mathrm{RY}(\delta)$ $\mathrm{RZ}(\delta)\mathrm{RX}(\delta)\mathrm{RY}(\delta)$ rotations on Data_Quantumbits
end for
$Values \leftarrow Measurement(Pauli - Z(Data_Quantumbits))$
$Labelled_Class \leftarrow softmax(Classical_Values)$
return

4 Result analysis

4.1 Experimental setup

The above experiment is performed using a high performance computer Param Shavak DLGPU (Ubuntu 18.04 LTS, 64 bit) with a frequency of 2.00 GHz × 40 and a RAM of 96 GB DDR4. It has Nvidia QUADRO RTX 5000 GPU with 16 GB of RAM. Python is used as the language of medium. NumPy, TensorFlow, Keras and Pennylane are some major libraries and tools which are used in our model. NumPy is used to generate a 1-D feature vector with the help of the extracted features from X-Ray images, tensorflow and keras libraries are used to build the MLDC-ANN. Here ANN classifier model has been trained for 50 epochs with learning rate of 0.0001 and ADAM optimizer.

For MLDC-QC, Python library Pennylane is used and qdevice is used as simulator. Qdepth taken here is 4 as classical features are transformed into encoded four qubits. Total number of parameters used here are 72 with a learning rate of 0.004 and from which 19 parameters are used for optimizing. We have used 3 CNOT Gates and 16 single qubit gates.

4.2 Performance analysis of MLDC on multiclass dataset(LDD)

We have used common evaluation metrics such as Accuracy, Precision and Recall to evaluate the performance of our model. The number of images predicted correctly are true positive (TP) and the number of images predicted wrong are false negative (FN). True Negative (TN) is the normal images predicted positive, and False Positive (FP) is a number of normal images predicted correctly [21]. Accuracy, measuring the proportion of correctly predicted instances, provides a high-level overview of performance, while recall emphasizes the ability to detect positive instances, crucial in scenarios with high consequences for false negatives. Precision, on the other hand, underscores the accuracy of positive predictions, vital in minimizing false positives. Each of these factors is indispensable in training and evaluating machine learning models, with

their selection and tuning depending on specific task requirements and dataset characteristics.

Figure 1 represents the overview of the methodology that has been carried out for this experiment. The feature extraction has been carried out using ResNet50, VGG16, MobileNet, Xception, DenseNet121, InceptionV3, InceptionResNetV2 and our proposed model MLDC-F. The extracted features from each of the pre-trained CNN models and our proposed model MLDC-F are fed as an input to existing standard classifiers such as K-Nearest Neighbors(KNN), Decision Tree (DT), Random forest (RF), Support Vector Machine (SVM), AdaBoost, Naive Bayes as well as our proposed classifiers MLDC-ANN and MLDC-QC for performance analysis.

Table 2 shows the output feature map and the total number of parameters for the pre-trained CNN models and proposed model MLDC. It can be observed that the size of the output feature map and the number of parameters for MLDC-F is predominantly lesser than the existing pre-trained CNN models.

Table 3 compares the accuracy of the proposed model with the standard classifiers KNN, RF, SVM, DT, AdaBoost, Naive Bayes and the proposed classifier MLDC-ANN and MLDC-QC. The accuracy was compared on the feature maps extracted using pre-trained CNN models (VGG16, ResNet50, MobileNet, Xception, DenseNet121, InceptionV3, InceptionResNetV2) and the proposed MLDC-F.It presents the following observations:

- Our proposed model MLDC achieved the maximum accuracy of 97.5% for MLDC-QC and 95.6% for MLDC-ANN among all the compared models.
- The accuracy of the DT and Naive Bayes classifier is less than 85%.
- The accuracy of our proposed classifiers outperforms RF, SVM and AdaBoost.
- The minimum accuracy achieved in MLDC-QC, MLDC-ANN, RF, SVM, DT and KNN is 89.8%, 90%, 86.4%, 88%, 80.6% and 85.1% respectively. The proposed classifier MLDC-ANN performs superior to the rest of the classifiers.
- Our proposed feature extractor MLDC-F gives better results in case of RF (93.6%), AdaBoost (94.5%), MLDC-ANN (95.6%) and MLDC-QC (97.5%).
- ResNet50 gives the accuracy of 94.5% on Adaboost classifier. However, the number of parameters and size of the output feature map is almost three times more than MLDC-F. The performance of this combination is slightly less than our proposed model(1.1% for MLDC-ANN).
- Figure 6 demonstrates the box-plot distribution of Table 3. The accuracy values of the proposed model follow a normal distribution. On the other hand, the remaining CNN models have skewed distributions.

Table 2 Feature vector and parameter of each model

	Output feature vector	Parameters
ResNet50 feature	2048	23,587,712
VGG 16 feature	512	14,714,688
MobileNet feature	1024	3,228,864
Xception feature	2048	20,861,480
DenseNet121 feature	1024	7,037,504
Inception V3 feature	2048	21,802,784
InceptionResNet V2 feature	1536	54,336,736
Proposed feature extractor	256	1,327,880

Table 4 compares the precision of the proposed model with the standard classifiers KNN, RF, SVM, DT, AdaBoost, Naive Bayes and the proposed classifier MLDC-ANN and MLDC-QC. The accuracy was compared on the feature maps extracted using pre-trained CNN models (VGG16, ResNet50, MobileNet, Xception, DenseNet121, InceptionV3, InceptionResNetV2) and the proposed MLDC-F.It presents the following observations:

- Our proposed MLDC-F on MLDC-ANN and MLDC-QC has achieved the maximum precision value of 96% and 98% respectively.
- DT, KNN and Naive Bayes precision values are lesser in comparison with other classifiers.
- The precision of our proposed MLDC-F outperforms SVM, RF and AdaBoost.
- The minimum precision achieved in the case of our proposed classifier (MLDC-ANN and MLDC-QC), RF, DT, Naive Bayes and KNN is 89%,90%, 85%, 75%, 76% and 87% respectively. The proposed classifier MLDC-QC performs superior to the rest of the classifiers.
- Our proposed feature extractor MLDC-F gives better results in the case of SVM (92%), AdaBoost (92%) and proposed classifiers MLDC-ANN (96%) and for MLDC-QC(98%).
- Figure 7 demonstrates the box-plot distribution of Table 4. The precision values of the proposed model follow a normal distribution. On the other hand, the remaining CNN models have skewed distributions.

Table 5 compares the recall of the proposed model MLDC with the standard classifiers KNN, RF, SVM, DT, AdaBoost, Naive Bayes and the proposed classifier MLDC-ANN and MLDC-QC. The accuracy was compared on the feature maps extracted using pre-trained CNN models (VGG16, ResNet50, MobileNet, Xception, DenseNet121,

Table 3 Accuracy metrics comparison of different classification techniques

	Random forest	Decision tree	KNN	Naive bayes	AdaBoost	SVM	MLDC-ANN	MLDC-QC
ResNet50	91	80	88.3	81.6	94.5	91.2	92	95.6
VGG	93.6	82.4	91.1	82	93	92	93	94.1
MobileNet	90.1	81	90	81.5	93.2	90.1	92	93.9
Xception	89.9	81.5	91.2	80.6	91.4	91.6	91	92.1
DenseNet121	90.4	80.6	89.5	80	92.5	94	90	90.8
lnceptionV3	91.2	79.9	85.4	81.2	92	89.4	91	93.5
lnceptionResNetV2	93.6	81.2	86.6	82.3	93.4	90	93.4	89.8
MLDC-F	86.4	80.6	85.1	83.4	91.59	88	**95.6**	**97.5**

Bold represents the performance of our proposed Feature Extraction and Classification models

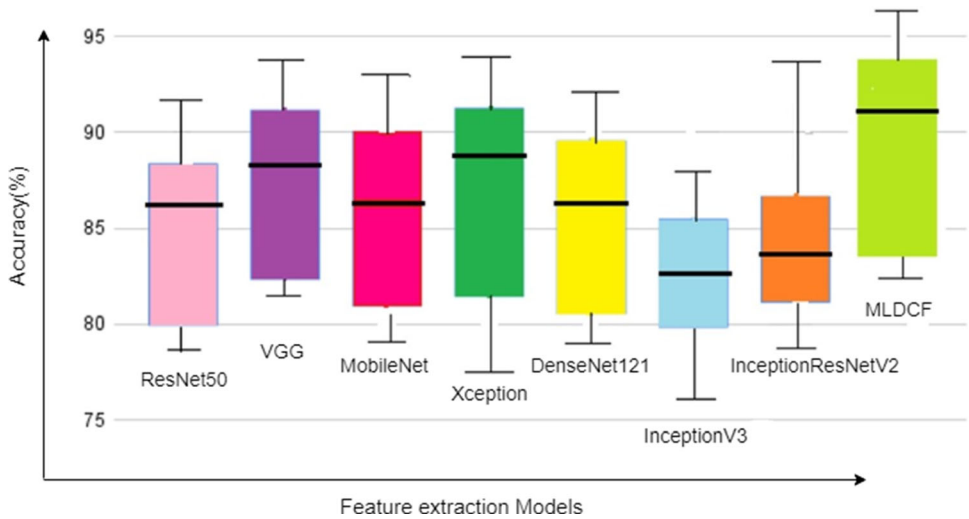

Fig. 6 Performance of the deep CNNs on classification in respect of accuracy metrics

Table 4 Precision metrics comparison of different classification techniques

	Random forest	Decision tree	KNN	Naive bayes	Ada boost	SVM	MLDC-ANN	MLDC-QC
ResNet50	91	82	89	95	94	96	95	96
VGG	94	82	93	80	93	92	92	93
MobileNet	92	81	95	80	94	95	93	92
Xception	90	81	91	81	91	94	95	92
DenseNet121	90	78	92	76	92	95	95	91
Inception V3	93	75	87	79	95	92	94	93
lnceptionResNetV2	94	79	87	80	94	92	89	90
MLDC-F	85	80	91	88	92	92	**96**	**98**

Bold represents the performance of our proposed Feature Extraction and Classification models

InceptionV3, InceptionResNetV2) and the proposed MLDC-F. It presents the following observations:

- Our proposed MLDC-F on MLDC-ANN and MLDC-QC, ResNet50 on MLDC-QC achieved the maximum recall value of 95.6%, 97.5% and 95.6% respectively.
- DT, KNN and Naive Bayes recall values are lesser in comparison with other classifiers.
- The recall of our proposed model MLDC-ANN, MLDC-QC and Adaboost outperforms SVM, RF.

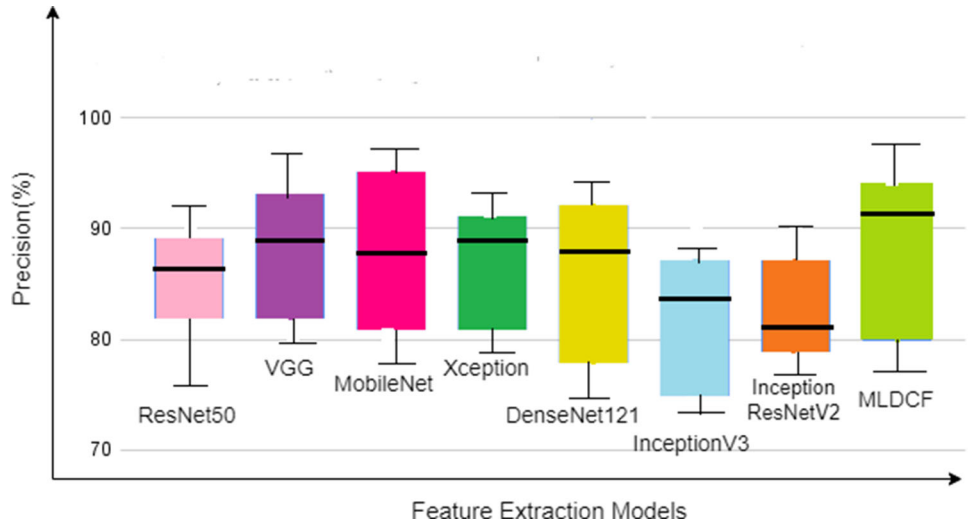

Fig. 7 Performance of the deep CNNs on classification in respect of precision metrics

- The minimum recall achieved in the case of our proposed classifier MLDC-ANN, MLDC-QC, RF, DT, Naive Bayes and KNN is 92%, 90%, 87%, 81%, 76% and 80% respectively. It can be observed that the proposed classifier MLDC-ANN performs superior to the rest of the classifiers.
- Our proposed feature extractor MLDC-F gives better results in the case of AdaBoost (91.59%) and proposed classifier MLDC-ANN (95.6%) and MLDC-QC (97.5%).
- Figure 8 demonstrates the box-plot distribution of Table 5. It can be observed that the recall values of the proposed model.

4.3 Performance analysis of MLDC on binary datasets

We have evaluated the performance of MLDC on binary dataset. We have considered four datasets. Dataset 1 and Dataset 2 corresponds to Covid-19 and non-Covid-19 X-Ray images. Dataset 3 presents Pneumonia and healthy X-Ray images of chest. Dataset 4 presents Tuberculosis and Normal X-ray images of chest. The dataset description for the above is given in Table 6.

Table 7 presents the comparison of existing models with our proposed MLDC-ANN and MLDC-QC on datasets (given in Table 6). It presents following observations:

- The models are compared on different parameters like Number of Trainable Parameters, Input Size, Learning Rate, Number of Layers, Batch Size, Optimizer and Accuracy.
- Narin et al. [24] with 9,921,451 trainable parameters has an accuracy value of 96% whereas our proposed model MLDC-ANN with 5,656,555 trainable parameters achieved an accuracy value of 95% on Dataset 1.
- On comparison with other models on Dataset 2 our proposed model MLDC-QC achieved maximum accuracy value of 97% on Dataset 2.
- loey et al. [25] has the minimum accuracy value of 85.2%.

Table 5 Recall metrics comparison of different classification techniques

	Random forest	Decision tree	KNN	Naive bayes	Ada boost	SVM	MLDC-ANN	MLDC-QC
ResNet50	92	81	88	93	94	92	95	96
VGG	92	87	91	87	93	90	95	94
MobileNet	91	83	89	83	93	91	94	93
Xception	89	82	91	82	92	93	93	92
DenseNet121	92	86	87	84	92	95	92	92
lnceptionV3	89	84	80	85	90	90	94	93
lnceptionResNetV2	93	84	86	84	93	92	92	90
MLDC-F	87	84	80	76	91	90	**95.2**	**97**

Bold represents the performance of our proposed Feature Extraction and Classification models

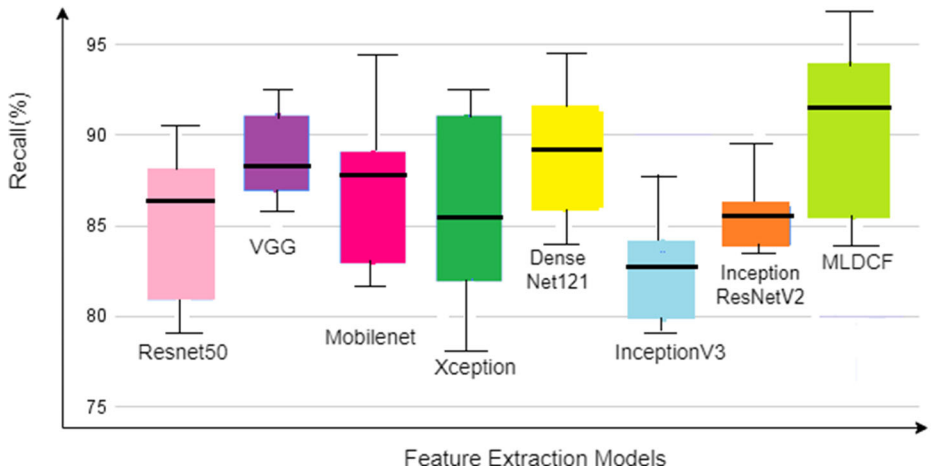

Fig. 8 Performance of the deep CNNs on classification in respect of recall metrics

Table 6 Dataset description

Serial No.	Dataset repositories	Dataset distribution		Total
		Positive	Negative	
Dataset 1	Actualmed COVID-19 chest X-ray dataset initiative [22]	58	127	185
Dataset 2	Covid-19 image data collection [23]	187	73	250
Dataset 3	Chest X-ray images (pneumonia) [19]	1572	1572	3144
Dataset 4	Tuberculosis chest X-ray database [20]	354	354	708

Table 7 Comparative analysis of proposed model with existing models on various datasets

Model name	Dataset	Input size	No. of trainable parameter	No. of layers	Learning rate	Optimizer	Batch size	Accuracy
COVID-19 dataset								
Shi et al. [26]	Dataset 1	224 × 224	2,246,125	18	0.0001	SGD	12	91%
Oh et al. [27]	Dataset 1	1024 × 1024	9,500,000	18	0.0001	ADAM	16	92.5%
Narin et al. [24]	Dataset 1	224 × 224	9,921,451	50	0.006	ADAM	3	96%
Coro-Net [28]	Dataset 1	224 × 224	33,915,436	71	0.0001	Sigmoid	10	89.6%
Deep -COVID [29]	Dataset 1	224 × 224	224,316	18	0.0001	ADAM	20	90%
Loey et al. [25]	Dataset 1	512 × 512	1,170,000	18	0.01	SGD	64	85.2%
MLDC-ANN	Dataset 1	224 × 224	5,656,555	21	0.0001	Sigmoid	32	95%
Brunese et al. [30]	Dataset 2	224 × 224	14,000,000	16	0.0001	Sigmoid	64	93.2%
MTSS-AAE [31]	Dataset 2	224 × 224	26,215,216	12	0.0001	SGD	32	96%
COVID-Net [32]	Dataset 2	480 × 480	2,121,806	21	0.03	Sigmoid	64	93.3%
MLDC-QC	Dataset 2	224 × 224	5,656,555	21	0.004	Sigmoid	32	97.5%
Pneumonia dataset								
kermany et al. [33]	Dataset 3	224 × 224	207,130	48	0.0001	ADAM	1000	92.8%
Babic et al. [34]	Dataset 3	64 × 64	78,785	9	0.0001	ADAM	10	95%
MLDC-ANN	Dataset 3	224 × 224	327,548	21	0.0001	Sigmoid	32	96%
MLDC-QC	Dataset 3	224 × 224	327,548	21	0.004	Sigmoid	32	98%
Tuberculosis dataset								
Rahman et al. [35]	Dataset 4	224 × 224	825,158	23	0.0001	SGD	32	98.6%
Dey et al. [36]	Dataset 4	224 × 224	23,000,000	23	0.0002	ADAM	8	99%
MLDC-ANN	Dataset 4	224 × 224	656,432	21	0.0001	Sigmoid	32	98%
MLDC-QC	Dataset 4	224 × 224	656,432	21	0.004	Sigmoid	32	98.2%

- Coro-Net [28] has the maximum number of trainable parameters i.e, 33,915,436 and has the highest number of layers i.e, 71 and is getting an accuracy of 89.6% on Dataset 1 which is less than our model MLDC-ANN.
- Kermany et al. [33] on Dataset 3 has the minimum accuracy value of 92.8%.
- Babic et al. [34] on Dataset 3 has an accuracy value of 95% by using only 78,785 number of trainable parameters.
- MLDC-ANN and MLDC-QC on Dataset 3 has an accuracy value of 96% and 98% respectively which is the highest among all.
- Rahman et al. [35] on Dataset 4 has the maximum accuracy of 98.6%.
- Dey et al. [36] shows the highest accuracy value of 99% on Dataset 4. The number of trainable parameters used in this paper are 23 Million.
- MLDC-ANN and MLDC-QC on Dataset 4 has an accuracy value of 98% and 98.2% respectively which is also close to that of the existing models.

Experimental results shows that the feature extractor module is better than other standard deep learning approaches (can be seen from Table 2). Similarly Tables 3, 4, 5 and 7 proves that the performance of ANN classifier outperformed the standard classifiers and further QC classifier is better than the ANN classifier model wr.t. the metrics Accuracy, Precision and Recall.

5 Conclusion

We investigated the use of deep learning algorithm, specifically a CNN algorithm and a quantum classifier, for multi-class classification of lung diseases using chest X-Ray images. Our research compared how well these two classifiers performed in correctly identifying and categorizing various lung diseases. Our findings demonstrated that both classifiers were highly accurate. The quantum classifier MLDC-QC accuracy was 97.5%, while the MLDC-ANN accuracy was 95.6%. These findings imply that quantum classifiers, among other deep learning algorithms, can be an effective tool for precise multi-class lung disease classification on chest X-Ray images. Additionally, our research revealed that the quantum classifier outperformed CNN in terms of speed and efficiency, suggesting that quantum classifiers may speed up the medical image analysis. To completely comprehend the potential of quantum classifiers in medical image analysis and their practical applications, more study is necessary.

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